AML Detection Is a Rare-event Network Problem

0.1019%

Full-data laundering rate

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Do account graph features help after accounting for imbalance and leakage risk?

Why Accuracy Fails

99.8981%

accuracy from predicting every full-data transaction as legitimate

Recall = 0
F1 = 0

Main metrics

Recall
F1
PR-AUC
Alerts per 10,000

The base rate makes accuracy look excellent even when the detector finds nothing.

The Model Needs Account Context, Not Just Rows

Transaction-level features

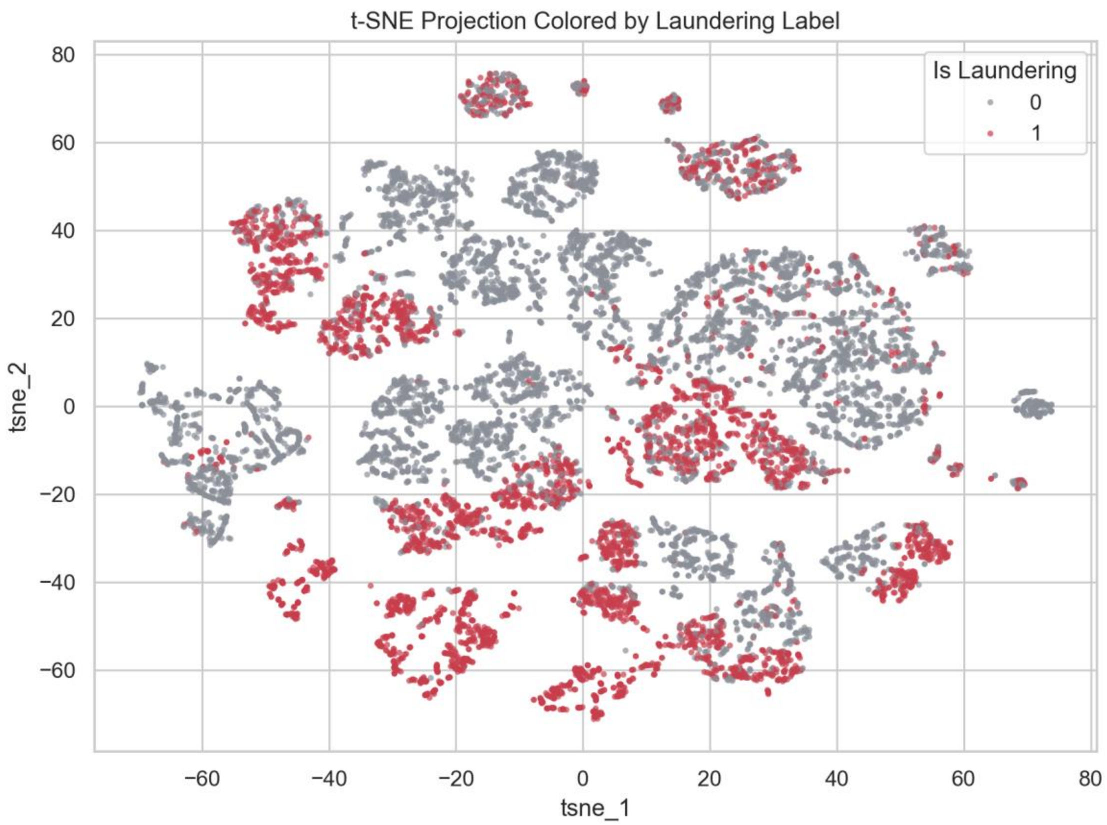
Amount and log amount
Currency and payment format
Hour and weekday
Same-bank and same-currency flags

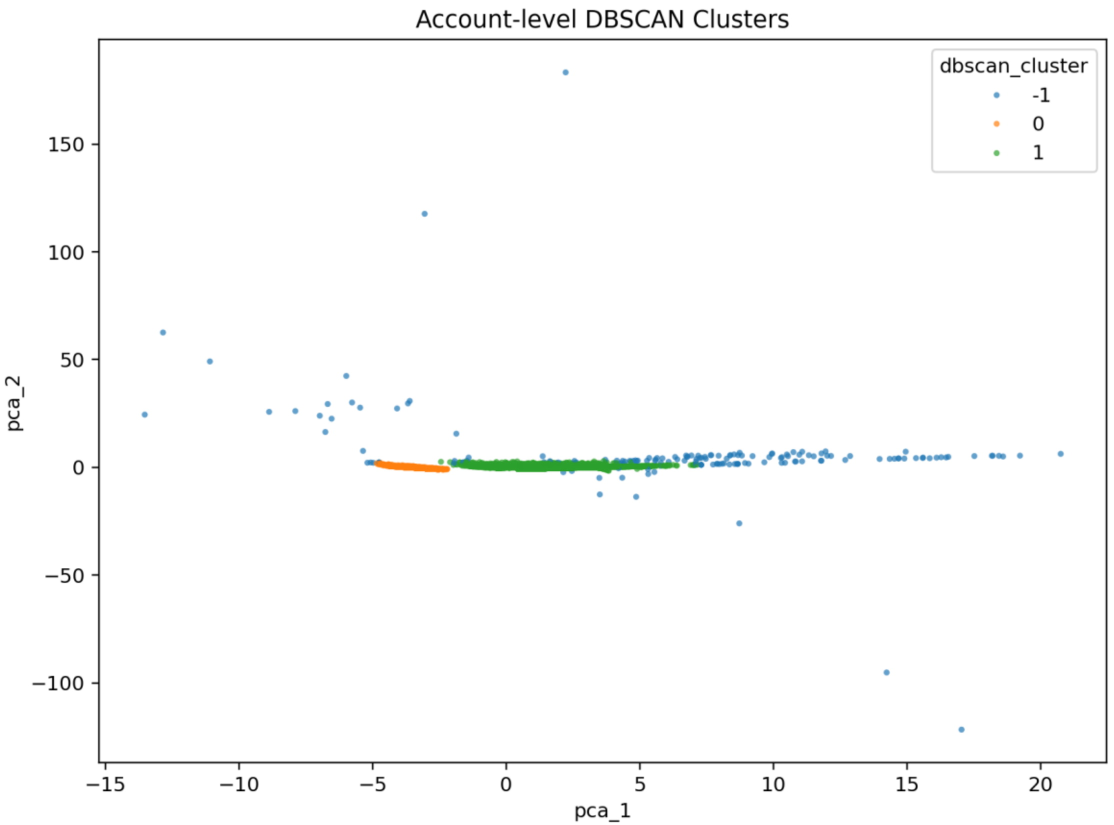
Account graph behavior

In-degree and out-degree
Unique counterparties
Total sent and received
PageRank
Fan-in, fan-out, scatter-gather proxies

The same transaction can look ordinary alone and suspicious in its account network.

Exploration Finds Local Risk Regions



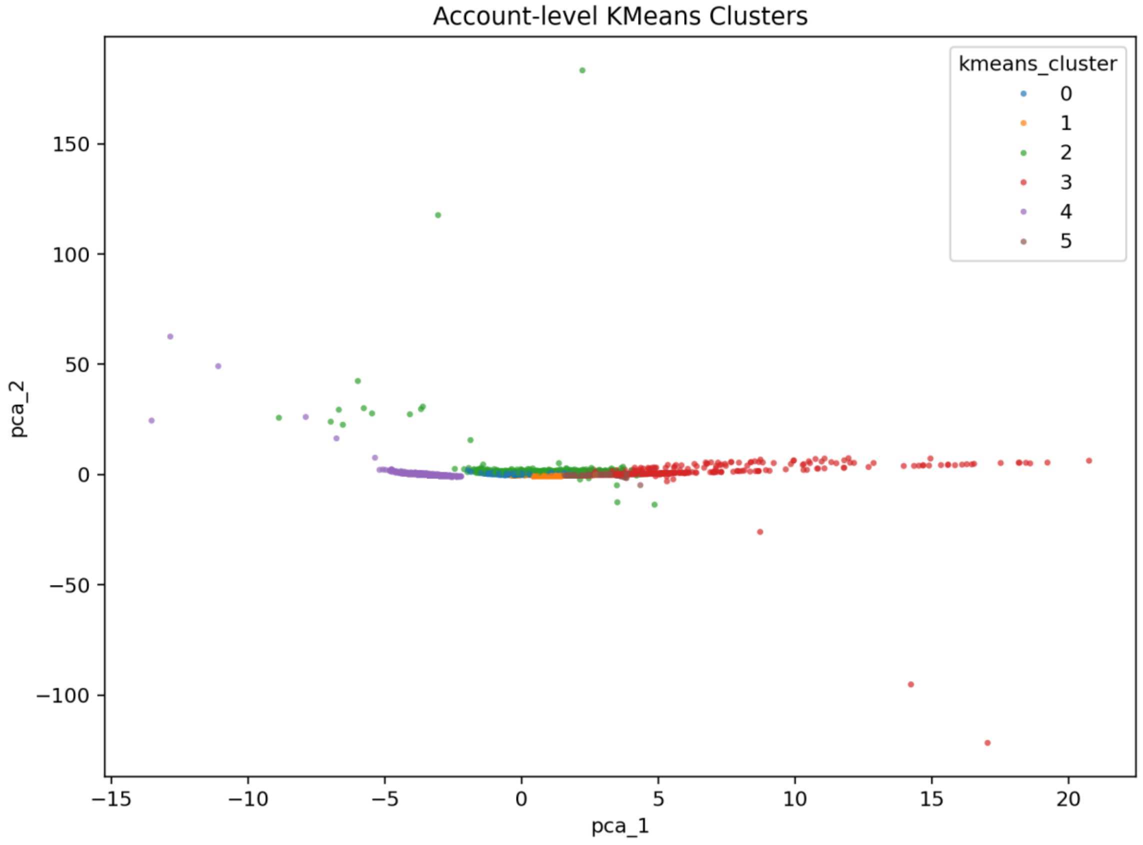


t-SNE: local neighborhoods with substantial overlap

DBSCAN: Silhouette 0.478 | NMI 0.001 | ARI -0.008

Clustering identifies investigation regions but cannot replace supervised prediction.

Clustering Quality Needs Cautious Interpretation



0.380

KMeans Silhouette

0.848

DBSCAN noise/outlier-group maximum risk

NMI and ARI remain near zero. Purity is inflated by class imbalance.

The high-risk DBSCAN noise/outlier group is a triage signal, while weak label agreement limits classification value.

Each Failure Changed the Next Experiment

01
Naive baseline

Accuracy cannot be the main AML metric; use recall, F1, ROC-AUC, and PR-AUC.

02
Linear baseline

Use it as an interpretable baseline, not as the final detector.

03
Single tree

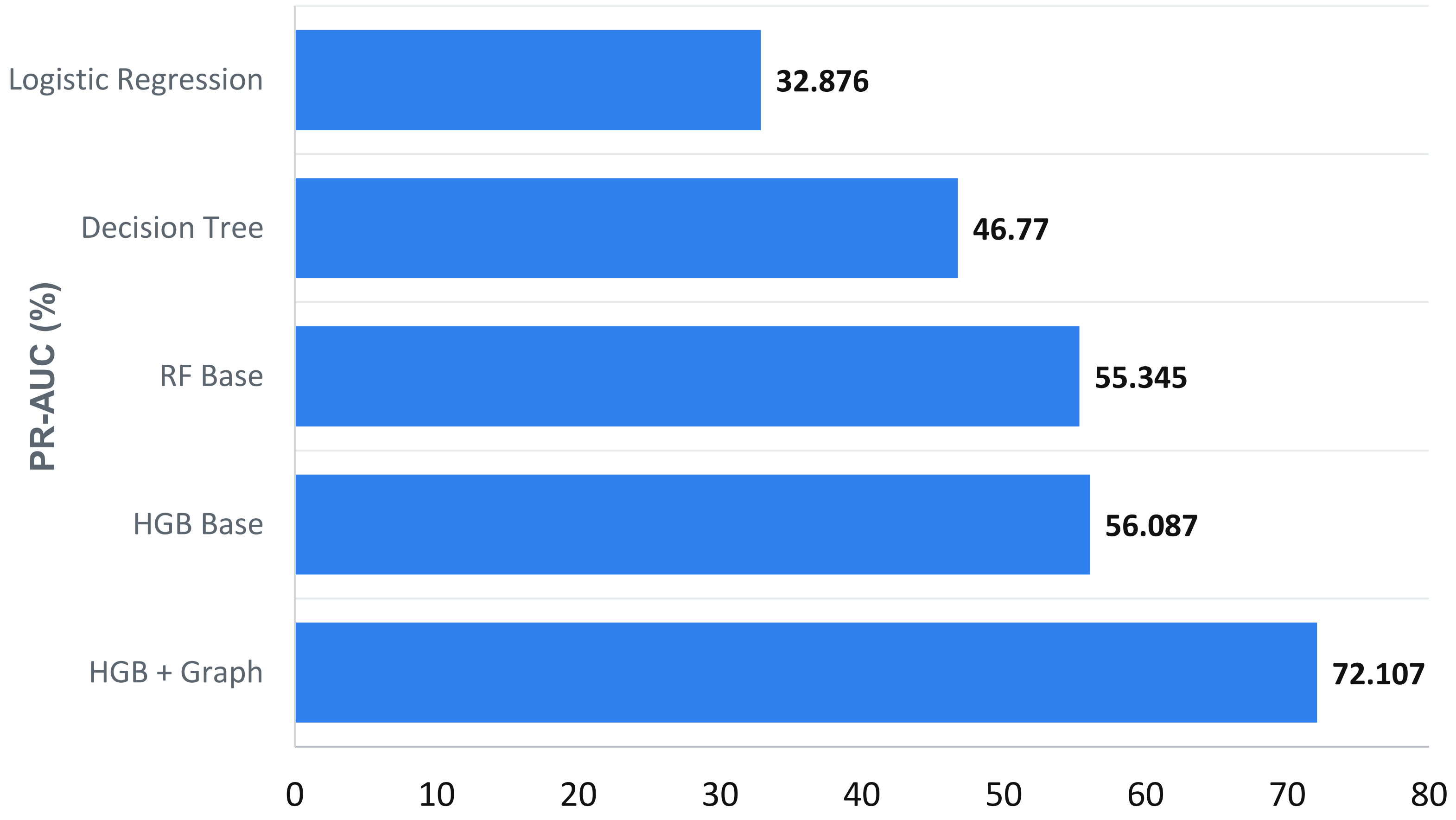
A single tree is useful for diagnosis, but ensemble models are more stable.

04
Temporal repair

A frozen graph snapshot failed; strict expanding history recovered a positive time-test gain.

The project moved from accuracy failure to cross-validation, ablation, and a leakage-safe temporal repair.

3-fold CV Confirms the Graph-aware HGB Lead



0.561 +/- 0.009

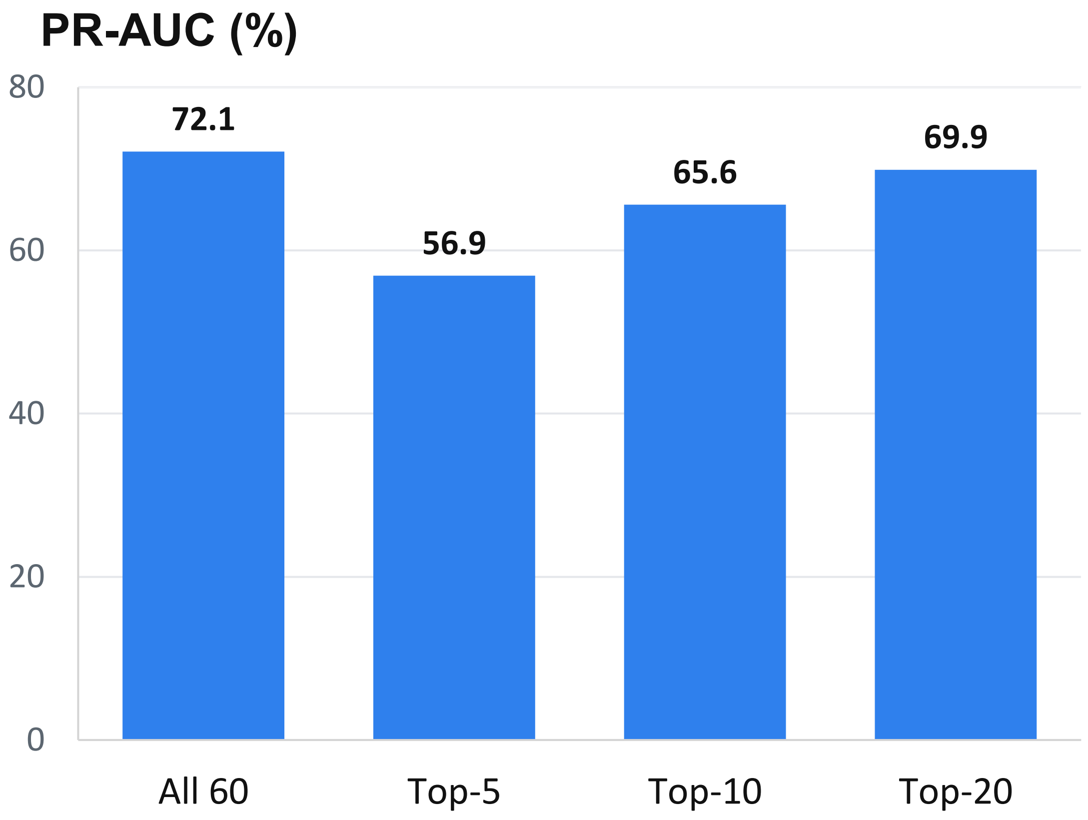
HGB Base CV

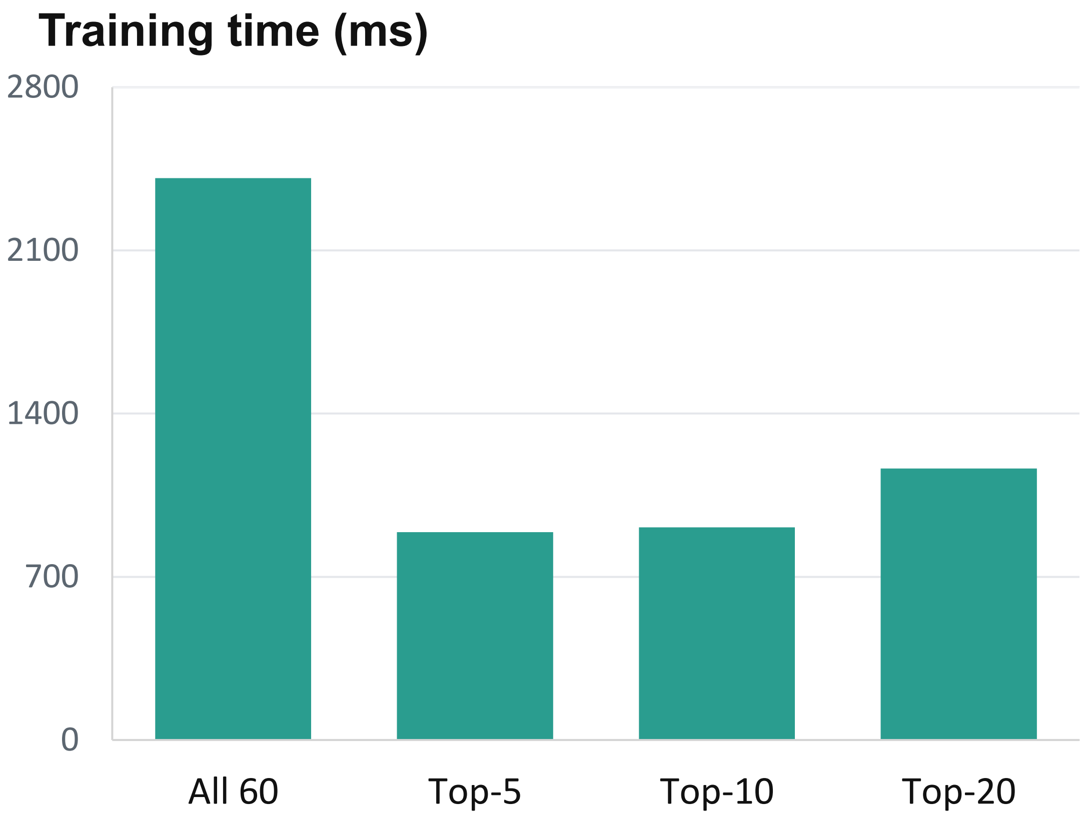
0.721 +/- 0.005

HGB + Graph CV

Graph-aware HGB leads all tuned candidates and remains stable across folds.

Top-20 MI Retains Most Performance with Less Work

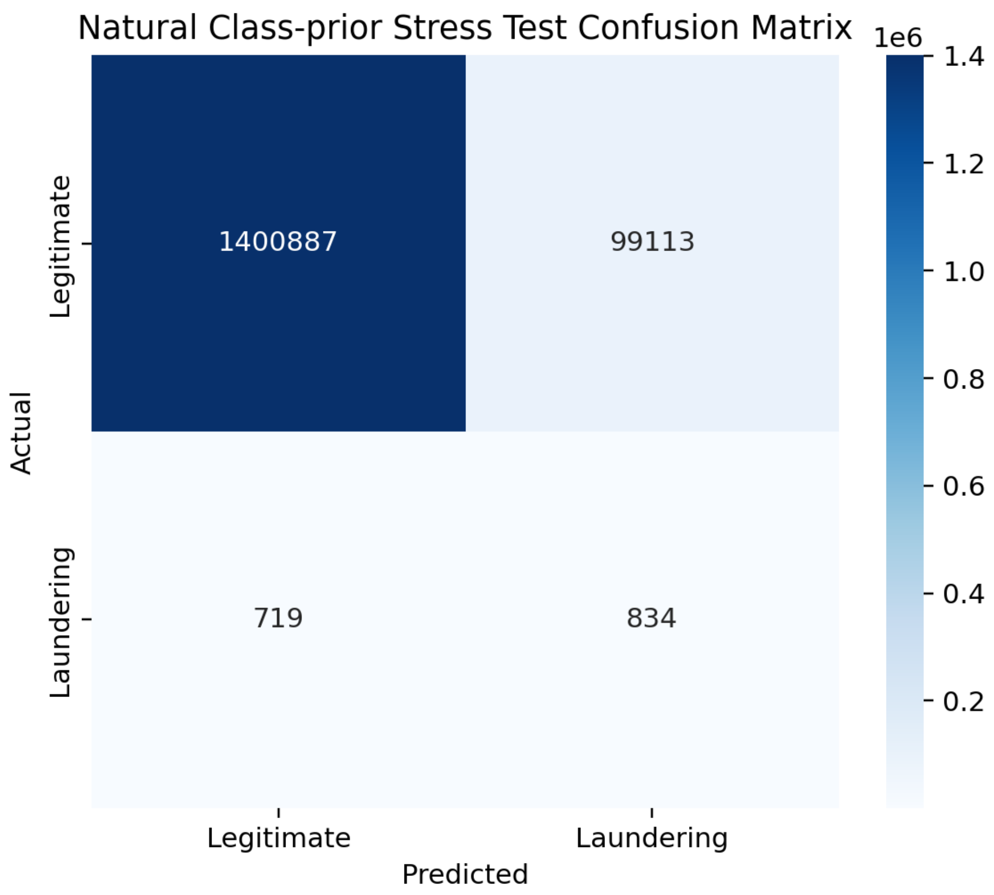




All 60: 2.41 s | Top-20: 1.16 s

Top-20 retains most sampled-test performance with 20 features and lower training time.

Natural Class Prior Turns Ranking into Alert Pressure



0.1034% Positive rate

0.83% Precision

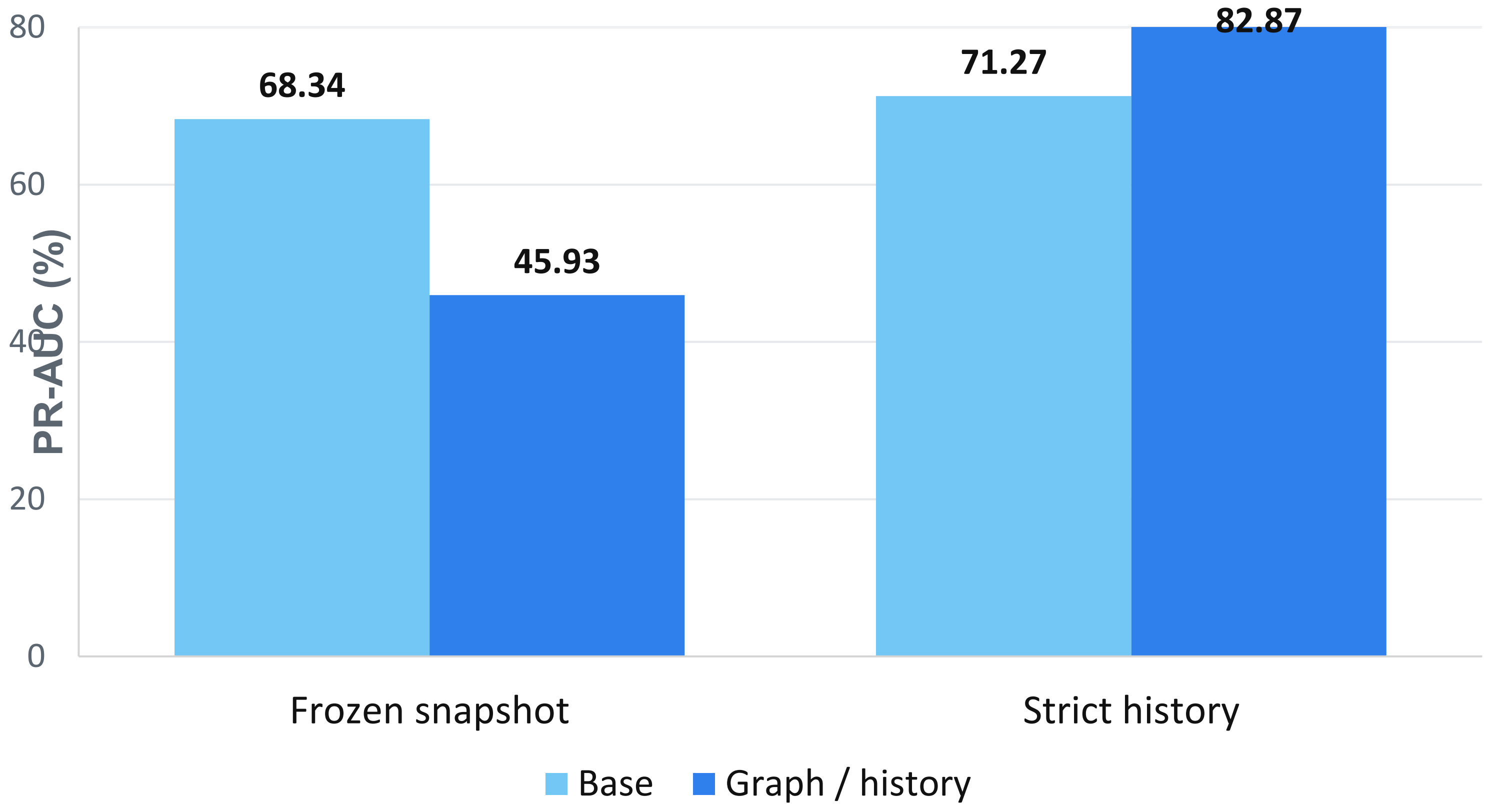
53.70% Recall

99,113 False positives

665.6 Alerts per 10,000

Good ranking performance does not automatically imply a manageable alert volume.

Strict History Repairs the Temporal Graph Failure



Frozen snapshot

Graph change -0.224 PR-AUC

Strict history

Graph gain +0.116 PR-AUC

Prior-only history improves time-test PR-AUC; paired 95% CI [+0.100, +0.130].